EXPERIMENT 12

Creating Reports and Dashboards for Data Visualization in Zoho Creator

Subject: No-Code Application Development

Aim

To design reports and dashboards that summarize leave requests and support managerial decision-making.

Objectives

- Create tabular and filtered reports from application records.
- Choose meaningful dashboard indicators and charts.
- Check correctness of aggregated values and role permissions.

Selected Application

Employee Leave Management app with a Leave Request form and Status values Pending, Approved and Rejected.

Requirements / Tools

- Existing Leave Request form with fictional data
- Report builder, filtering and dashboard/page features
- Sample departments and approval statuses

No-Code Connection

A no-code report builder converts saved records into filtered lists and charts without hand-writing the whole front end. Dashboard correctness depends on data quality and filter scope.

Illustrative Business Scenario

Create a status summary for leave requests and test whether totals equal the rows in the source report.

Experimental Procedure

Follow these steps in the designated simulator, browser or visual app builder. Menu locations and available features may vary by product version and account plan.

1. Open the application’s report builder and generate a list report for leave submissions.
2. Display useful columns: Employee Name, Department, Leave Type, Start Date, End Date and Status.
3. Create filtered report views such as Pending Requests and My Requests.
4. Build a dashboard page with counts grouped by approval status and department.
5. Add an appropriate chart, for example a bar chart for requests by department.
6. Compare displayed totals with source records and verify each role sees only permitted data.

No-Code Application Perspective

- Identify the user action and the visual interface element involved (form, page, workflow or report).
- Trace the data movement from input to processing to stored record, report or notification.
- Identify configuration that replaces traditional hand-written code and note where scripting might still be required.
- Check whether the experiment can be repeated using sample records and documented settings.

Evidence to Collect

- Screenshot of the configured screen, topology, tool or application.
- A short observation table showing input, action and output.
- One successful test and one failed or exceptional test with expected behavior.

Sample dataset (illustrative)

Request	Department	Status
L001	Production	Pending
L002	HR	Approved
L003	Production	Approved
L004	Quality	Rejected
L005	HR	Pending

Dashboard indicators from sample

Indicator	Value
Total requests	5
Pending	2
Approved	2
Rejected	1
HR requests	2

Interpretation / Analysis

- Filters enable managers to focus on requests requiring action.
- Status counts must reconcile with the underlying dataset; a chart should always have clear category labels.
- Restricted leave details require secure record-level visibility.
- The sample counts are computed from the illustrative dataset, not from a live Zoho application.

Evaluation Against a No-Code Solution

- Ease of creation: determine how much is configured visually rather than programmed.
- Data and integration: identify where records, connections or services are created and managed.
- Governance: consider role permissions, traceability and safe data handling.
- Limitations: identify any feature that may require paid access, external services or custom logic.

Verification / Test Cases

Suggested Test Record

Test	Input / action	Expected observation
Normal test	Valid sample values and supported action	Expected successful output
Boundary test	Missing value, invalid input or empty data	Validation or clear empty state
Access test	Unapproved user attempts restricted action	Access limited according to role
Repeatability	Re-run same valid example	Consistent output or explain differences

Observation Notes

During the practical session, replace the illustrative outcomes with actual observed results. Include the software/platform name, version if known, screenshots, and any errors encountered.

Results

The proposed dashboard presents five total requests: two pending, two approved and one rejected, demonstrating how summaries can support operational review.

Conclusion

Well-designed reports and dashboards convert form data into actionable business information while preserving accuracy and access control.

No-Code Learning Outcome

This experiment links the technical concept to a practical no-code application development workflow. It highlights visual configuration, user experience, data handling and safe validation rather than treating the exercise as conventional full-code software development.

Precautions

- Use only fictional sample records for classroom demonstrations; avoid exposing passwords, tokens, employee information or real contact details.
- Record your own screenshots, observed outputs and deviations after executing the experiment. The examples below are models, not verified live results.
- Check access rights before sharing or publishing an app, report, workflow or externally reachable service.